Unicast Routing Protocols & Algorithms

Coverage: Link-State Routing & Dijkstra's Algorithm, Distance-Vector & Bellman-Ford, Count-to-Infinity, RIP, OSPF & BGP-4

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. LS vs DV Routing

- Link-State: Floods LSAs; Dijkstra shortest path algorithm (OSPF).
- Distance-Vector: Distributed Bellman-Ford. Exchanging vectors with neighbors (RIP).

2. Internet Protocols

- RIP (DV, Hop count max 15, UDP 520) | OSPF (LS, Bandwidth metric, IP 89) | BGP-4 (Path Vector, TCP 179).

■ KTU University High-Yield Exam Questions

- Q: Explain Dijkstra's shortest path algorithm with a worked table. (9 Marks)